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22F-3277
BCS-3C
LAB 8

LAB 8
(Total Marks 15)

Task 1: Write assembly language code to display your name and roll number on screen by using following instructions:

- Display your name using blinking of foreground character and green component of background color.
- Display your roll number using blinking of foreground character red component of foreground color.

```
[org 0x0100]
```

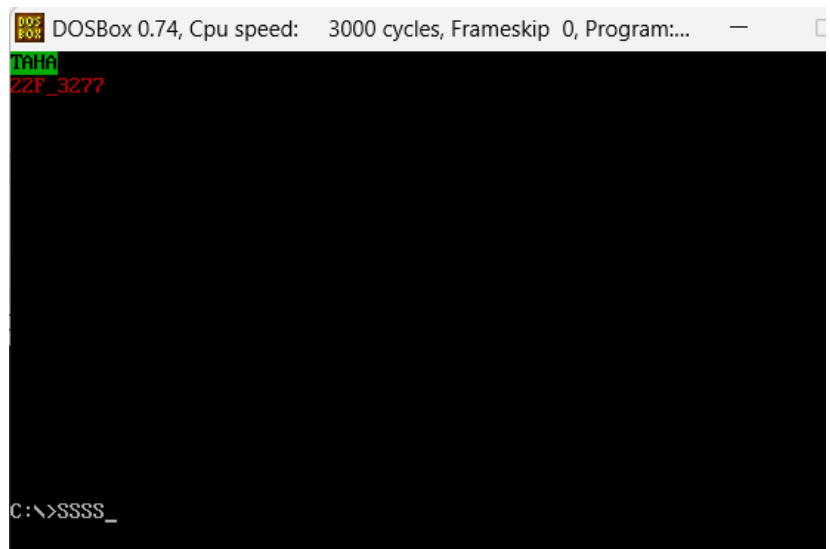
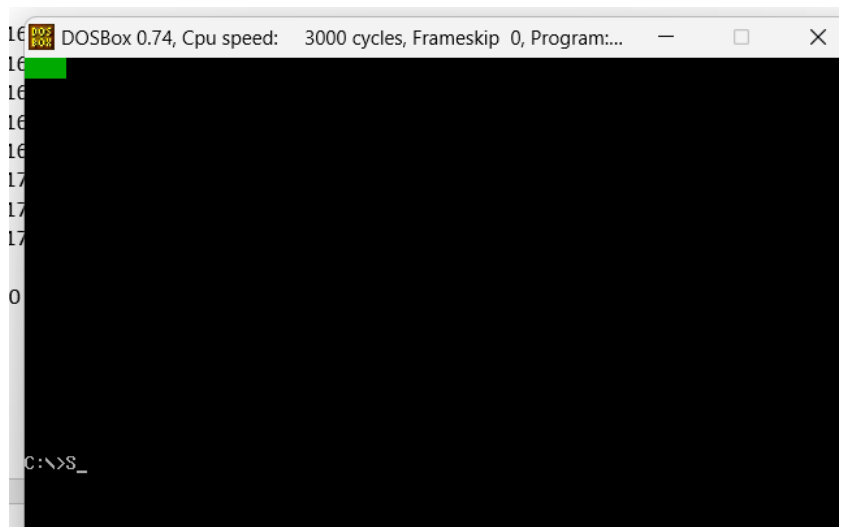
```
mov ax, 0xb800  
mov es, ax  
mov di, 0
```

```
nextchar:  
    mov word [es:di], 0x0720  
    add di, 2  
    cmp di, 4000  
    jne nextchar
```

```
mov word[es:0], 0xA054  
mov word[es:2], 0xA041  
mov word[es:4], 0xA048  
mov word[es:6], 0xA041
```

```
mov word[es:160], 0x8432  
mov word[es:162], 0x8432  
mov word[es:164], 0x8446  
mov word[es:166], 0x845f  
mov word[es:168], 0x8433  
mov word[es:170], 0x8432  
mov word[es:172], 0x8437  
mov word[es:174], 0x8437
```

```
mov ax, 0x4c00  
int 0x21
```



Task 2: Write assembly language code to display first alphabetic of your name and last digit of your roll number with following instructions:

- Design both characters in form of '*'.
- Use blinking of foreground character and blue component of background color.

```
[org 0x0100]
```

```
mov ax, 0xb800
mov es, ax
mov di, 0
```

```
nextchar:
```

```
    mov word [es:di], 0x0720
    add di, 2
    cmp di, 4000
    jne nextchar
```

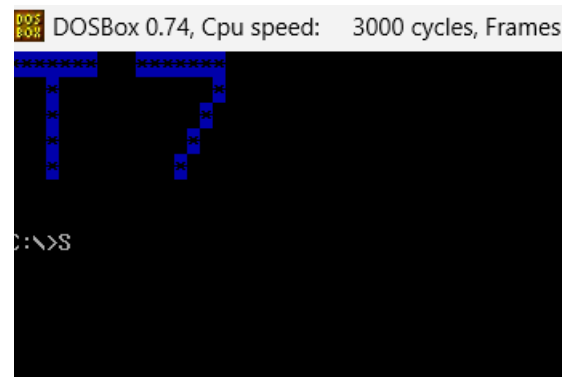
```
mov word[es:0], 0x902A
mov word[es:2], 0x902A
mov word[es:4], 0x902A
mov word[es:6], 0x902A
mov word[es:8], 0x902A
mov word[es:10], 0x902A
mov word[es:12], 0x902A
```

```
mov word[es:20], 0x902A
mov word[es:22], 0x902A
mov word[es:24], 0x902A
mov word[es:26], 0x902A
mov word[es:28], 0x902A
mov word[es:30], 0x902A
mov word[es:32], 0x902A
```

```
mov word[es:166], 0x902A
mov word[es:326], 0x902A
mov word[es:486], 0x902A
mov word[es:646], 0x902A
```

```
mov word[es:192], 0x902A
mov word[es:350], 0x902A
mov word[es:508], 0x902A
mov word[es:666], 0x902A
```

```
mov ax, 0x4c00
int 0x21
```



Task 3: Write a code which will take position and length of snake and then print it.

- Position Input: you will take x and y position.
- Calculate its location using formula $\text{location} = (\text{y position} * 80 + \text{x position}) * 2$
- Taking all parameters from the stack
- You should clear the screen first and then proceed to print it.

```
org 0x100
```

```
jmp start
```

```
clrscr:
```

```
    push es
```

```
    push ax
```

```
    push di
```

```
    mov ax, 0xb800
```

```
    mov es, ax
```

```
    mov di, 0
```

```
nextloc:
```

```
    mov word [es:di], 0x0720
```

```
    add di, 2
```

```
    cmp di, 4000
```

```
    jne nextloc
```

```
    pop di
```

```
    pop ax
```

```
    pop es
```

```
    ret
```

```
snakeMaker:
```

```
    push bp
```

```
    mov bp, sp
```

```
    push es
```

```
    push ax
```

```
    push cx
```

```
    push si
```

```
    push di
```

```
    mov ax, 0xb800
```

```
    mov es, ax
```

```
    mov di, 0
```

```
    mov ax, 80
```

```
    imul ax, [bp+6]
```

```
    add ax, [bp+8]
```

```
    shl ax, 1
```

```
    mov cx, [bp+4]
```

```
    mov di, ax
```

```
l1:
```

```
mov word [es:di], 0x042A
add di,2
dec cx
cmp cx,0
jne l1
```

```
pop di
pop si
pop cx
pop ax
pop es
pop bp
ret 4
```

start:

```
call clrscr
```

```
mov ax, 20 ;x position
push ax
mov ax, 10 ;y position
push ax
mov ax, 15 ;length
push ax
```

```
call snakeMaker
```

```
mov ax,0x4c00
```

```
int 0x21
```

